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APPLICATIONS OF OR IN DISASTER RELIEF OPERATIONS

Embedded analytics: improving decision support for humanitarian logistics operations

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Abstract Analytical techniques continue to advance in efficacy, as well as complexity. However, it is sometimes unrealistic to employ complex analyses during time-constrained humanitarian disaster operations. We propose that simple, embedded analytics tools can provide an effective and practical means toward managing humanitarian operations. In this paper, we demonstrate a real-world application of our technique in a patient evacuation context. This paper contributes to literature and practice by showing how simple analytic methods and open-source imagery tools can offer significant value to the humanitarian operations literature. The application also highlights some challenges to drawing a clear picture from disparate data sources in the humanitarian operations domain.

Keywords Business analytics · Big data analytics · Humanitarian logistics · Decision tree · Embedded analytics · Data science

1 Introduction

Humanitarian operations are significant international concerns, and have both political and economic implications (Kent 2004). Nearly 80% of time and resources spent on humanitarian

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operations are consumed by logistical efforts (Bonney 2005; Trunick 2005). As such, the success of humanitarian operations is directly attributable to how efficienct and effective decision makers are in getting the necessary people and supplies to the right place.

Humanitarian logistics is defined as the "process of planning, implementing, and controlling the efficient, cost-effective flow and storage of goods and materials, as well as related information, from the point of origin to the point of consumption for the purpose of alleviating the suffering of vulnerable people. The function encompasses a range of activities, including preparedness, planning, procurement, transport, warehousing, tracking and tracing, and customs clearance." (Thomas and Kopczak 2005, p. 2).

Scientific research supporting humanitarian logistics has received increased attention in recent years (Banomyong et al. 2017; Anaya-Arenas et al. 2014). When considering humanitarian logistics, the need to understand a particular structure, model, or set of variables can be an incredibly challenging and complex process (Overstreet et al. 2011). Although analytic efforts to improve humanitarian efforts have been made, the research area remains analytically immature as the methodologies employed remain centrally focused around a few common optimization techniques where the computational burdens may prevent them from being used in actual disasters (Özdamar and Ertem 2015). This lack of robust analytic methodologies and apparent infeasibility can drive a disconnect between the perceived relevance of analytics and decision-making for practitioners (Hazen et al. 2016).

Furthermore, organizations are moving towards a self-service analytic mentality. Instead of a few analysts generating analytics or reports, organizations are pushing the generation and consumption of analytics closer to frontline supply chain employees so that decisions around supply chain operations can be made more quickly (Dinsmore 2016). Providing data and analytics at the point of need is exceedingly important in humanitarian logistics where the analytics-to-decision cycle time can determine life or death outcomes. As such, as the number and impact of disasters have increased, there has been a growing interest from scholars in an attempt to self-service analytic tools and solutions (Coles et al. 2017). Indeed, humanitarian logisticians require analytic capabilities that enable humanitarian logisticians to independently perform data discovery, visualization, and build dashboards to aid and improve decision-making. However, creating such capabilities has its challenges. Those challenges are due to self-service analytic capabilities requiring:

1. A balance in the complexity of the analytic methodology, as non-analyst consumers will need to apply and interpret the capability and its results;
2. An operational deployment structure; and
3. Data to be centralized, although the current Big Data environment produces many disparate data sources making the data collection and processing challenging.

This research proposes a decision support tool that leverages a simple and intuitive analytic technique that is embedded in an open source platform to provide self-service data discovery and visualization capabilities for humanitarian logisticians. The purpose of this paper is to showcase how this embedded analytic method can offer significant value to humanitarian logistics efforts that require collecting, analyzing, and leveraging data resources to solve emerging problems. Through demonstrating a real-world application of our technique, this paper makes the following contributions.

Contribution 1 We show how a simple analytic method and open-source imagery tools can offer significant value to the humanitarian operations literature.

Contribution 2 Our development efforts to produce these embedded tools illustrate the challenges of drawing a clear picture from disparate data sources in the humanitarian operations domain.

The rest of this paper proceeds as follows. First, we provide a brief background on surrounding humanitarian logistics literature, and the decision tree tool employed in this research. Then, Sect. 3 describes the problem, data, analyses, and results. We close with a discussion of the findings, and some concluding remarks.

2 Background

2.1 Humanitarian logistics

To date, academic research has primarily focused on two aspects of the humanitarian operation lifecycle: pre-disaster and post-disaster. These aspects are further subdivided into three distinct phases: preparedness (pre-disaster), response (post-disaster), and recovery phases (post-disaster) (cf. Özdamar and Ertem 2015; Kovács and Spens 2009). First, research on the pre-disaster preparedness phase addresses improving preparedness and mitigation strategies via *preparedness plans* (cf. Kovács and Spens 2009; Van Wassenhove 2006), *risk-prevention actions* (cf. Maon et al. 2009; Perry 2007), *pre-positioning planning and modeling* (cf. Hale and Moberg 2005; Balcik and Beamon 2008; Widener and Horner 2011; Duran et al. 2011; Mete and Zabinsky 2010), and *scenario analysis* (cf. Mete and Zabinsky 2010; Balcik and Beamon 2008; Chang et al. 2007). Second, research on the post-disaster response phase concentrates on *vehicle routing* to satisfy relief delivery demand (cf. de Angelis et al. 2007; Shen et al. 2009; Berkoune et al. 2012) *evacuation procedures/traffic flow management* (cf. Chui and Zheng 2007; Mastrogiannidou et al. 2009; Song et al. 2009; Cui et al. 2014) and *casualty transportation* (cf. Haghani and Oh 1996; Özdamar et al. 2004; Özdamar 2011). Third, research on the post-disaster recovery phase focuses on *recovery logistics* that is concerned with proper damage assessment, effective reverse logistics systems for debris disposal and recycling, and infrastructure rebuilding (cf. Allen et al. 1993; Feng and Wang 2003; Liberatore et al. 2014; Özdamar and Ertem 2015).

A key differentiator between the abovementioned phases is the focus on speed and cost reduction. Historically, research on the preparedness and recovery phases focused more on cost minimization whereas research on the response phase focused on the speed of the response (Tomasini and Van Wassenhove 2009). With urgency and speed as a main driver in the response phase, the analysis-to-decision lead time reduction becomes an important area of consideration. Industry-based research suggests that the elapsed time to complete an organizational analysis-to-execution process provides a value-add of only 3–5% (Handfield and Nichols 1999). Thus, in humanitarian logistics, reducing the analysis-to-execution process time not only has the potential to save lives but it can also minimize time spent on processes that add little to no value added.

Although the preparedness phase emphasizes urgency, once the operation has been set up, roles have been defined, and there is better visibility about the process to assist beneficiaries, the response supply chain starts to resemble a normal business supply chain, which seeks to balance speed and cost of response. (Tomasini and Van Wassenhove 2009). A concrete example would be the choice of transportation mode. In the initial response phase, organizations may choose to fly in all aid with little concern for cost. However, to supply the same response support in subsequent weeks they may seek more efficient options for supplying

response aid, such as coordinating with regional air support, using road or sea delivery, or working through local actors to distribute the aid in the communities. Thus, an appropriate balance between speed and cost should be made in humanitarian logistics research.

2.2 Analytic approaches

Supply chain and operations researchers have employed analytic techniques such as optimization, simulation, and network models to understand characteristics, options, and behaviors of humanitarian logistics (Özdamar and Ertem 2015). Techniques such as classical vehicle routing (Berkoune et al. 2012; Shen et al. 2009; Nolz et al. 2011) and linear and integer programming have been used to optimize travel time, cost, and demand. Dynamic network flow has been used to reduce unmet demand and improve equity of demand provided (Özdamar et al. 2004; Zhang et al. 2012; Afshar and Haghani 2012). The choice of analytic technique is often a function of the problem and the analytic maturity of the organization (De Oliveira et al. 2012). However, as Özdamar and Ertem (2015) point out, much of the analytic research remains centrally focused around a few common optimization techniques whose "computational burdens might prevent them from being used in actual disasters" (p. 63). Furthermore, Coles et al. (2017) highlight the burgeoning gap between humanitarian logistics research and practictioner needs in guided decision-making in the field. Thus, a wider variety of analytic approaches would benefit the humanitarian community. The decision tree is one such approach given that it is computationally feasible, simple to deploy, relatively intuitive, and is fundamentally about guiding decision-making. Yet, humanitarian logistics research has not leveraged decision trees.

Due to their ability to break down complex decision-making processes into a collection of simpler decisions, decision tree classifiers have been used successfully in many diverse areas (Safavian and Landgrebe 1991). Decision trees create classifiers that provide insight and understanding into the predictive structure of the data rather than focusing only on predictive accuracy. Tree-based methods involve stratifying or segmenting the predictor space in to a number of simple regions. This partitioning suggests that not all decision points of the tree must be explicitly considered in order to arrive at an optimal decision. Furthermore, the partitioning enables trees to have a high degree of flexibility that improves performance. The significant benefits of decision trees are in their computational simplicity, intuitive alignment to the decision-making process, and interpretability of results. This study deploys a decision tree approach to aid in humanitarian logistics decision-making.

It is very difficult, if not impossible, to implement solution procedures without a proper information system that operationalizes and embeds analytic models in a user-friendly interface (Özdamar and Ertem 2015). Unfortunately, the use of technology is still not prevalent in humanitarian logistics and few researchers have attempted to embed their analytic modeling into humanitarian operational processes equipped with geographical information systems (GIS), maps, real-time data, and a user-friendly interface (Overstreet et al. 2011; Özdamar and Ertem 2015).

Operationalizing and embedding analytics is about integrating actionable insights into systems and business processes used to make decisions (Leventhal and Langdell 2013). The goal is to integrate analytics into operational systems so that users will be able to access historical data and perform analytics on those data while executing transactions and making decisions (Shankararaman and Gottipati 2015). As the number and impact of disasters have increased, there has been a growing need for the research community to provide such tools (Coles et al. 2017). However, this often requires capital, and capital for information technology personnel and infrastructure at humanitarian organizations has historically been limited

(Oloruntoba and Gray 2006; Thomas and Mizushima 2005). Thus, research needs to illustrate how alternative technology approaches such as open source software can be leveraged to advance analytic capabilities (Boehmke and Hazen 2017). The solution presented in this study illustrates such embedment in open source technology.

2.3 Data disparity

Organizations can now collect massive amounts of transactional data, which can create both opportunities (Wang et al. 2016; Dubey et al. 2016; Papadopoulos et al. 2017) and problems (Hazen et al. 2014). The opportunities reside in the fact that big data analytic approaches have recently gained traction and are creating greater diversity in approaching a very complex set of problems (i.e. Fadiya et al. 2014; Burns 2015; Prasad et al. 2016). This is giving supply chain and OR research a more robust perspective (Waller and Fawcett 2013; Hazen et al. 2016).

However, a major problem associated with big data are the challenges presented when attempting to integrate disparate data sources. Each data source can contain a combination of data elements pertaining to customer transactions, inventory management, store-based video feeds, advertising and customer relations, customer preferences and sentiments, sales order management, and financial data. To provide self-service analytic capabilities, organizations must first orchestrate the process of collecting, storing, and processing these data feeds so they can be fed to the embedded analytic platform. Unfortunately, collecting, storing, and processing this data footprint creates significant challenges for organizations (Chen and Zhang 2014; Kambatla et al. 2014). As a result, humanitarian logisticians are potentially face with being "data rich", yet "information poor". This research highlights some of the challenges of drawing a clear picture from disparate data sources in the humanitarian operations domain.

3 Problem description and analysis

The principal purpose of this research is to provide a self-service analytic tool such that humanitarian logisticians can perform a data-focused comparison of various alternatives for patient evacuation. The emphasis fixates on deploying this self-service analytic and decision-making tool via an open source platform to maximize accessibility and minimize cost of an appropriate analytic solution for humanitarian operations. In doing so, this approach could ensure that practitioners can control the data discovery, visualization, and decision assessment process. Such control entails how to allocate scarce resources to their most efficient use for aeromedical evacuation. The self-service analytic approach employed in this study illustrates the problem through the following elements:

1. *Logistic map* nodes representing the places of pick-up, delivery or connection, and main links characterized by distance, round trip time, safety, and cost;
2. *Priority* the urgency of patient movement;
3. *Response time* the quickness of patient pick-up and delivery based on vehicle capabilities;
4. *Air transport vehicles* several types are considered, characterized by capacity, variable and fixed costs, availability, and in each node of the logistic map;
5. *Financial elements* the cost incurred for operations; and
6. *Additional criteria* such as regulatory concerns, additional training requirements for the mission, and competing demand of resources, among others.

The embedded analytic decision tool is designed for use at the operational level, when the organization in charge of the patient evacuation has to act expeditiously and in the best possible way. The objective of the embedded analytic decision tool is to show planners where the majority of aeromedical evacuation requirements exist and their current capability of aeromedical evacuation assets. Providing such clarity should allow planners to analyze potential sites according to their ability to provide medical coverage. More specifically, the tool enables planners to more quickly analyze their options for patient evacuation by (1) determining the preferred air transport vehicle to be used for aid distribution and (2) visually identifying gaps in evacuation capabilities.

3.1 Problem context

Since 2001, the United States Department of Defense (DoD) moved more than 150,000 patients in Iraq and Afghanistan (USTC 2015). Ranging from the aircraft design and dispatch centers to the alliances and partnerships with other nations and civilian corporations, the DoD has the capability to move large numbers of patients affected by calamities worldwide. Over the last 14 years, the DoD built medical treatment facilities in Iraq and Afghanistan to improve patient movement in theater. The infrastructure development activities by the DoD created a network of medical support that provided surgical care and stabilization in theater. This infrastructure also allowed the establishment of aeromedical evacuation channel missions that ensured patient movements occurred within the appropriate timeframes as specified by DoD regulation (Department of Defense 2012).

Despite the infrastructure improvements in the Middle East, the current construct of patient movement within Africa does not support the military to the same level. Africa is vast continent, consisting of 11.7 million square miles and approximately 56 countries. Each of these countries has various degrees of stability, economic prosperity, terrorist activities, and amenability toward the US. The DoD relies on multiple small-scale operations to pursue its interests in promoting a stabilized Africa. As a result, the DoD employs a small number of troops, spread over a very large area, with limited access to medical or aeromedical evacuation support (Whitlock 2012). The only base with a surgical medical capability in Africa is Camp Lemonnier, located in Djibouti (Schmidt and Johnson 2005).

In the current patient movement system, each military service component provides its own patient movement from point of injury to initial stabilization (Department of Defense 1996). The aeromedical evacuation system then moves stabilized patients from one medical treatment facility to another. In many countries, the DoD relies on the local medical systems for this support. However, due to a lack of adequate medical infrastructure in Africa, the DoD provides its own support in that region (Mullan et al. 2011). Given the sparse nature of troops, the complicated political environment, and the high cost to supply medical assets in theater, the DoD does not typically provide surgical level medical support to anyone outside the Camp Lemonnier catchment area. The inability to treat patients onsite tends to result in the generation of a Priority or Urgent Patient Movement Request (PMR) when members are ill or wounded. Further complicating matters, the Air Force does not staff, train, or equip the aeromedical evacuation system to move unstable patients. Therefore, the inability to stabilize and hold patients within Africa places unwarranted pressure on the aeromedical evacuation system to move patients in an unsafe manner (Department of Defense 2012).

Given that the closest military aeromedical evacuation support is located at Ramstein Air Base, Germany, it is difficult to move a Priority patient within the 24-h requirement. Furthermore, the distance also makes it nearly impossible to move an Urgent patient with military aeromedical evacuation assets. The vast distances, coupled with the high cost to support

few troops, and the poor state of the local medical infrastructure poses an obvious problem for the traditional construct of patient movement planning and execution. To compensate for this lack of medical and aeromedical evacuation support, the DoD relies on civilian air ambulance contracts to fill the gap in service. However, these contracts are expensive and have limited use during contingency operations. Compounding the issue, little analysis exists on the cost, incidence, distribution, causes, and requirements for aeromedical evacuation in Africa. This lack of awareness translates to a gap in preparation and planning that is necessary for executing adequate medical support in Africa.

3.2 Data

The data required for developing this self-service analytic capability reside in numerous disparate sources.

- Data regarding patients' locations, movement dates/times, precedencies (e.g. Routine, Priority, or Urgent), diagnoses, and demographics are located in the U.S. Transportation Command (USTRANSCOM) Regulating and Command and Control Evacuation System (TRAC2ES) database, which is controlled by the USTRANSCOM Surgeon General and maintained by the Global Patient Movement Integration Center. The TRAC2ES database captures every PMR made to USTRANSCOM since 2001.
- Data regarding U.S. Mobility Air Force missions, including aircraft type and take-off, landing, and ground times, requires data from the Global Decision Support System (GDSS), which is also controlled by USTRANSCOM.
- Data regarding U.S. Air Force flying costs, such as supplies, fuel, and maintenance and repair costs, require access to the Air Force Instruction 65-503 (AFCAIG 2015) database.
- Data regarding the cost of each civilian air ambulance movement as quoted by the firm International SOS requires access to the civilian air ambulance database, which is controlled by Theater Patient Movement Requirements Center-Europe.
- Data regarding civilian air ambulance dates and origins of civilian air ambulances reside in homegrown databases developed by Theater Patient Movement Requirements Center-Europe.
- Data regarding crew numbers, per diem, specialty pay, and Combat Zone Tax Exclusion for the U.S. Air Force missions are compiled from individual operations manuals and online sources.

Nearly all these data are access-restricted and require coordination and approval to be granted access. Once required access was granted, these disparate data sources required substantial cleaning, organizing, and transformation to allow for data merging and further analysis. For example, the raw TRAC2ES data are provided in separate yearly spreadsheets. These spreadsheets were merged and then filtered for only those records representing U.S. Africa Command missions in which the patient was not well enough to travel without medical supervision. This effort was further complicated by a change in codes and naming conventions, which represent these conditions, between 2010 and 2014. Once the TRAC2ES data was combined and filtered, it was merged with similarly filtered GDSS data. However, in this data merge, record disconnects were identified. For example, one mission identified in GDSS, but not in TRAC2ES, originated from an island with an African ICAO but fell under the jurisdiction of U.S. Pacific Command. Consequently, an iterative process of elimination, record review, and cross-referencing between the two databases was required to successfully reconcile a set of aeromedical evacuation missions listed in both GDSS and TRAC2ES.

Once the patient movement and corresponding mobility missions were reconciled and combined, an economic evaluation was required to determine the avoidable cost of each mission.[1] To compute the avoidable cost of each military flight, we first used the GDSS data to determine missions' total time duration. Total time duration is the elapsed time between the flight from the aircraft origin to the pick-up location and the flight from the pick-up location to the final destination. We then computed the avoidable costs by segmenting them into four categories (Horder 2003): *flight operations, maintenance, depreciation,* and *user charges.*

Flight operations include general crew and special medical crew expenses, fuel and oil. For *maintenance*, we include labor (if contracted), repairs, and material. We ignored *depreciation* since the market value of an aircraft depreciates primarily with time. Therefore, aircraft usage is only a concern if it is significantly different from an average aircraft (Conklin and deDecker 2016). *User charges* are the fees charged by governments and airports for using their facilities or airspace (Horder 2003). These fees include charges for landing, passengers, security, infrastructure, parking, environment, terminal usage, fly-over, and taxes. We identified and calculated all costs from the Air Force Instruction 65-503 dataset. Due to the unorthodox format of the data (Wickham 2014) we had to manually incorporate the cost rates for the GDSS data.

For U.S. Air Force missions, we used the number of personnel aboard each aircraft to calculate the cost of per diem, special pay, and Combat Zone Tax Exclusion. To determine the number of personnel, we used the base crews for each aircraft as outlined in individual operating procedures (C-21 Operations Procedures 2010; C-17 Operations Procedures 2011; C-130 Operations Procedures 2012; Aeromedical Evacuation Operations Procedures 2014). We then determined if the crew required augmentation based on the calculated total mission time. If the total mission time was more than the basic flight day, we considered it an augmented crew unless there was enough ground time for crew rest, in which case, we considered it a basic crew. Additionally, we added the appropriate medical crewmembers if the mission required a Critical Care Air Transport Team. Lastly, per diem rates were obtained through the Per Diem Rates Query Tool on defensetravel.dod.mil and special pay locations and Combat Zone Tax Exclusions were derived from the DoD 7000.14-R regulation. With the aforementioned data elements, we computed the total personnel pay and savings.

3.3 Decision tree

The medical evacuation planners' incumbent process for determining which aircraft to use is rather straightforward. They examine the time requirement to move the patient, and then they give military airlift the right of first refusal. If the military cannot support the mission within the given timeframe, then International SOS is contracted to perform it. This simple decision tree does not consider factors such as training, cost, risk, or opportunity cost. Furthermore, the

[1] Using guidance from Martin (1993), we address only the truly avoidable costs incurred when choosing between organic and contracted patient evacuation flights, instead of the absolute cost of flying an aeromedical evacuation mission. Hence, we assume that the US Department of Defense (DoD) will not reallocate committed resources as a result of contracting aeromedical evacuation missions. We also assume the DoD will not completely privatize aeromedical evacuation and will maintain an organic aeromedical evacuation capability. When the DoD determines to "consume" an additional aeromedical evacuation mission, they do not need to increase the resources supplied by a full "mission's worth" if they have excess capacity that is already committed to the mission (Kaplan and Cooper 1998). For example, the pilots, aeromedical evacuation crews, aircraft, runways, and control towers are committed resources and will not change with an additional mission. Therefore the only additional costs incurred are the truly variable costs such as fuel, maintenance, and supplies. In the same manner, aeromedical evacuation missions, like all services, cannot be inventoried and have no residual value.

DoD's use of fully allocated overhead costs can foster poor incentives and resulting actions. On one hand, the cost structure incentivizes the overuse of assets in an effort to decrease the cost per mission. On the other hand, any inflated costs render the DoD uncompetitive with the market. A proper decision tree would attempt to balance these incentives in an effort to maximize the effectiveness of the entire system and allocate scarce resources in a manner that ensures each patient receives the required care at the lowest possible cost to the system.

Our decision tree analytic tool identifies the aircraft that minimizes the cost of aeromedical evacuation missions, while meeting the criteria specified by the attending doctor and DoD regulations. Within these tools, we include duty day, crew rest requirements, aircraft speed, aircraft size, aircraft origin, and cost per flying hour as constraints. Administrative and diplomatic factors were deemed out of scope and not considered further.

The decision tree algorithms developed and illustrated below walk the user through the necessary steps to identify the most economical aircraft choice, by asking the user to make choices in a series of alternatives. Inputs considered in the decision tree's development include contingent environment, precedence category, distance from Ramstein Air base, number of patients injured, necessary training requirements, avoidable cost, and "no-stop" requirements. The tool's highest priority was to meet the patient care standards as designated by their respective *Priority/Urgent* or *Routine* movement classification. Financial implications are secondary to patient care; the avoidable cost became the dominant discriminator after a planner meets the patient's care requirements. To minimize the decision tree's size, we designed the decision tool to eliminate possibilities quickly. This decision tree identifies the optimal aircraft choice for a "normal" movement. However, other factors affect aircraft choice such as aircraft availability, weather, political boundaries, and patient severity. Therefore, this and any similar tool's recommendation should be considered in the proper context. Internal validity was established through an iterative process that consisted of coordinating with senior leaders and front-line logisticians that are involved in this decision-making to ensure each decision point aligns to operational needs and concerns.

3.3.1 Priority/urgent decision tree

1. What is the Precedence of the Patient Movement Request?
   a. If Urgent or Priority → Go to step 2
   b. If Routine → Use the Routine Tree (found on subsequent pages)
2. Is the patient located in a contingent environment?
   a. If Yes →
      i. Is the patient less than 1850 miles from Ramstein Air Base (C-130 duty day range)? → Go to step 3
      ii. Is the patient more than 1850 miles from Ramstein? → Use a C-17[2]
   b. If No → Go to step 3
3. How far is the patient located from Ramstein Air Base?
   a. If less than 1450 miles (C-21 duty day range)? → Go to step 4
   b. If between 1450 miles and 1850 miles? → Use a C-130
   c. If more than 1850 miles? → Go to step 6

[2] While this is the optimal choice, in this scenario, the system is incapable of reaching Urgent patients located further than 2500 miles or Priority patients further than 3900 from Ramstein Air Base.

4. Does the 86th Aeromedical Evacuation Squadron need to conduct training?
   a. If Yes → Use a C-130
   b. If No or Do Not Know → Go to step 5
5. How many patients need to be moved?
   a. If less than or equal to one stretcher (litter) and two ambulatory patients → Use a C-21
   b. If more than one litter and two ambulatory patients → Use a C-130
6. Did the Validating Flight Surgeon mandate a No-Stop requirement?
   a. If Yes → Go to step 7
   b. If No → Go to step 8
7. How far is the patient located from Ramstein Air Base?
   a. If less than 2300 miles (civilian air ambulance fuel range)? → Go to step 8
   b. If between 2300 miles and 3900 miles? → Use a C-17[3]
   c. If more than 3900 miles? → Go to step 8[4]
8. How many patients need to be moved?
   a. If less than or equal to two litter and one Ambulatory patients? → Use a Civilian Air Ambulance
   b. If more than two litter and one Ambulatory patients? → Use a C-17

Routine Decision Tree (use only for Routine Patient Movements):

1. Did the Validating Flight Surgeon mandate a No-Stop requirement?
   a. If Yes → Go to step 2
   b. If No → Go to step 3
2. How far is the patient located from Ramstein Air Base?
   a. If less than 2300 miles (C-21 fuel range)? → Go to step 3
   b. If between 2300 miles and 2600 miles (C-130 fuel range)? → Use a C-130
   c. If between 2600 miles and 3,900 miles? → Use a C-17
   d. If more than 3900 miles? → Go to step 6
3. How many patients need to be moved?
   a. If less than or equal to one litter and two Ambulatory patients? → Go to step 4
   b. If more than one litter and two Ambulatory patients → Use a C-130
4. Is there space available on military aircraft already scheduled to move from Ramstein to the patient's location and back?
   a. If Yes → Use space-a
   b. If No → Go to step 5
5. Does the 86th Aeromedical Evacuation Squadron need to conduct training?
   a. If Yes → Use a C-130

[3] While this is the optimal choice, in this scenario, the system is incapable of reaching Urgent patients located further than 2500 miles from Ramstein Air Base.

[4] Civilian air ambulances go to South Africa if located more than 3900 miles from Ramstein.

b. If No → Use a C-21

6. Is the patient located in a contingent environment?

   a. If Yes → Use a C-17
   b. If No → Go to step 7

7. How many patients need to be moved?

   a. If ess than or equal to two litter and one Ambulatory patients → Civilian air ambulance to South Africa
   b. If more than two litter and one Ambulatory patients → Use a C-17

3.4 Map application

Google's My Map (Google Inc. 2015) was used to create the self-service analytic tool and provide practitioners control of the data discovery, visualization, and decision tree assessment process. The tool leverages 10 map layers on which we plot aircraft locations, ranges, areas of responsibility, costs, and patient movement data. The first step was to plot the basing location of the aircraft and the corresponding operating areas for the civilian air ambulances. We assume that the civilian air ambulance closest to a patient's origin is the primary responder unless the patient is returning to South Africa. In this case, the civilian air ambulance from Johannesburg is the primary responder. Four map layers respectively represent the operating regions for four civilian air ambulance locations. The military airlift layer covers all of Africa. The second step was to plot the ranges for each aircraft on a cost basis.

Using an airlift choice's average speed and avoidable cost per flight-hour, we calculated the range that each civilian air ambulance location could support for different price intervals and plotted these ranges on the map. For military airlift, we computed the maximum range of each aircraft type based on its average air speed, ground time, augmented duty day, and standard alert times.

We used the final two layers to plot all the Urgent and Priority patient movements represented in our data. These layers allow the user to see actual data points, reinforcing the tool's accuracy.

3.5 Map tool example

Operationally, this tool provides a quick method of determining the preferred and lowest cost aircraft for aeromedical evacuation with the given set of decision tree criteria. Figure 1 is an image of the map tool with all layers activated. This figure shows that the cost of aeromedical evacuation varies widely depending upon location and aircraft choice. For an example, the user can use the highlighted white tear-shaped marker, which represents an Urgent movement from Burkina Faso. Immediately, you see this patient is beyond the range of the C-130 and C-21 Urgent capability, very close to the C-17 boundary, and well within the Nigerian civilian air ambulance area of responsibility.

Eliminating all other layers (Fig. 2) we see the patient movement would cost over $188 K with a C-17 and between $100 and $150 K (yellow lines) for a civilian air ambulance. Given that this is a single patient and that the civilian air ambulance is more capable of moving the patient within the prescribed timeframe, this tool recommends the use of a civilian air ambulance. However, this location is beyond the "non-stop" range of the civilian air ambulances, so the plane requires a fuel stop en route. To choose the appropriate aircraft, the decision maker must weigh this information against the quicker response time and lower cost of the civilian

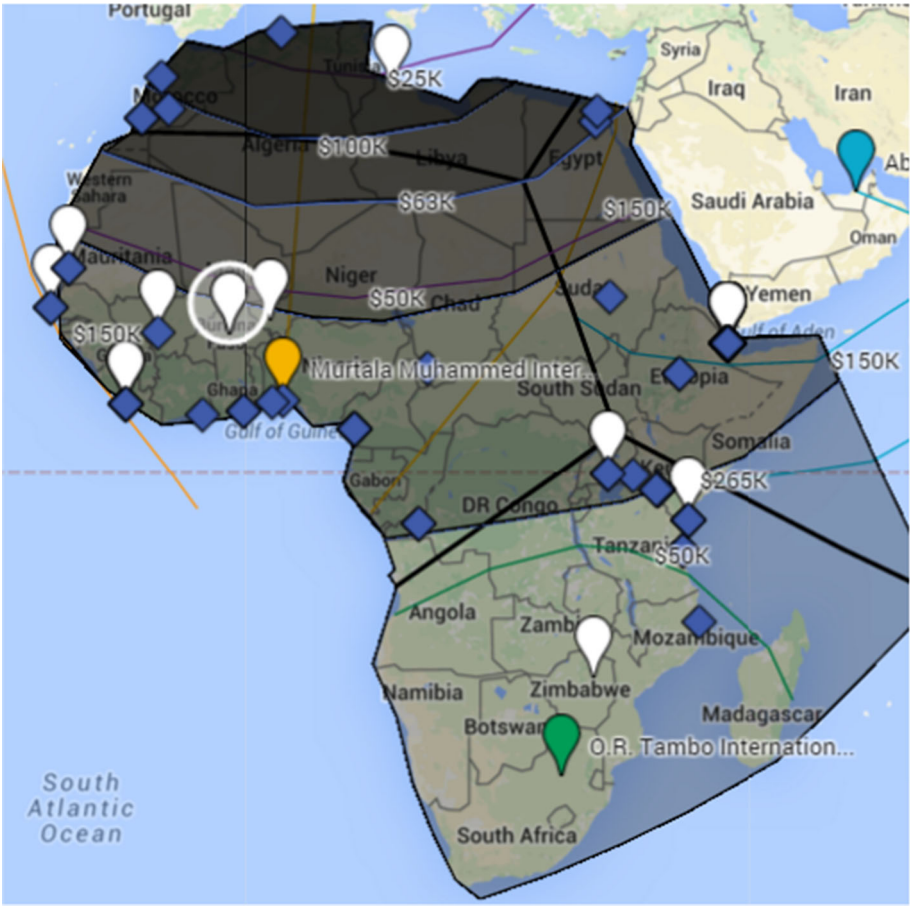

Fig. 1 Aeromedical evacuation map tool with all layers activated

air ambulances. Since each patient situation differs, the Map Tool defers to the Validating Flight Surgeon and The Aeromedical Evacuation Control Team to make the final decision. Although the tool does not provide the definitive answer in every case, this illustrates how it provides quick information to help support the decision maker.

The tool also provides information on past moves by selecting the corresponding marker. Figure 3 shows that a civilian air ambulance moved this patient over a 12 h period with a destination of Landstuhl Regional Medical Center and a cost of $129 K. This capability allows the user to perform exploratory data analysis in a visual environment to better understand historical performance.

Providing this data discovery capability allowed planners to identify coverage and capability gaps. For example, with this tool, it became easy to see that the South African civilian air ambulance responds to an Urgent request out of Zimbabwe for $50 K. At the same time, a MilAir (military) flight to the same location cost over $265 K and cannot meet the necessary timelines. If the civilian air ambulance capability is unavailable, it is impossible to meet the requirements with the current MilAir system.

Furthermore, planners could visualize the limited fuel ranges of certain aircraft and the impact of that limitation. The tool made it easy to see that many patient locations (to include

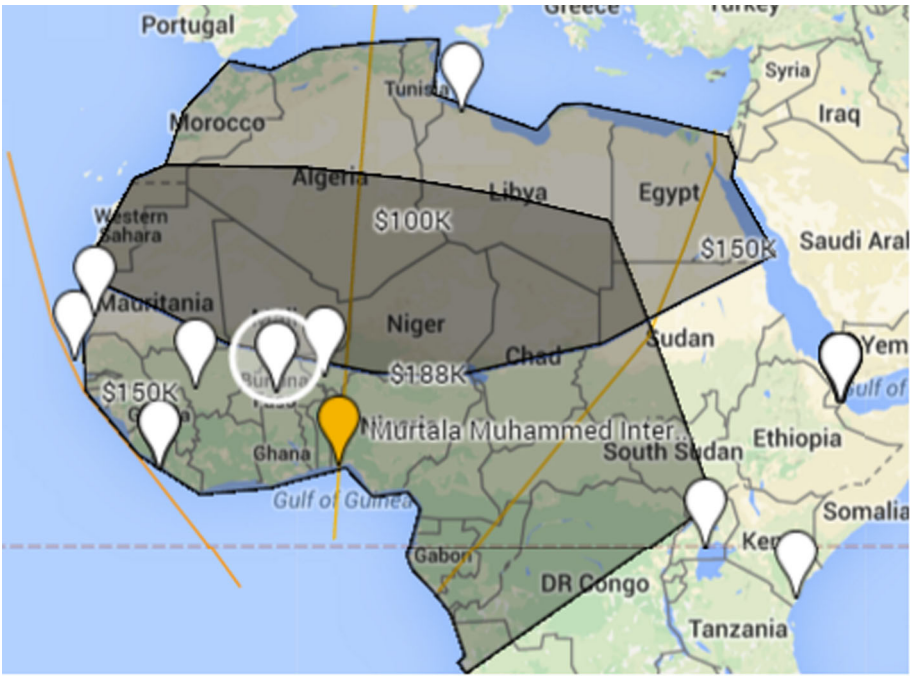

Fig. 2 Western civilian air ambulance AOR and Urgent C-17 layers

Djibouti) fell outside most aircraft fuel ranges. As a result, patients must either endure additional stressors that occur during takeoff and landing to refuel or the DoD must use a C-17, its most expensive form of patient movement. This fact stresses the limits of aeromedical evacuation capability in Africa, even with the civilian air ambulance capability.

3.6 Results

Our self-service analytic tool provides planners with control of the data discovery, visualization, and decision assessment process regarding how to allocate scarce resources in Africa. The new analytic tool provides a more robust decision framework than the previous decision approach, and reduced the analysis-to-decision time. In addition, because the analytic tool was designed using open source technology, there were no expenses incurred for software or licenses.

Our tool also provided planners quantitative and visual insight on deficiencies with AFRICOM medical assets. The visual illustrations quickly made it apparent to senior leadership that the DoD does not have the organic ability to move urgent or critical patients in many locations without significant planning and resource dedication. Furthermore, even with civilian air ambulance support across the entire African continent, several limitations exist that may jeopardize patients.

Lastly, with our tool fully developed, we compared 170 historical missions with our decision tree's respective recommendation for aircraft choice and associated cost implications. We found that 46 of the 170 missions (27%) used an inferior aircraft choice. Within these 46 missions, an inferior aircraft was chosen 24% of the time for Urgent missions, 16% of the time for Priority missions, and 64% of the time for Routine missions. These results were interesting since Routine missions have the longest planning horizon and should represent the

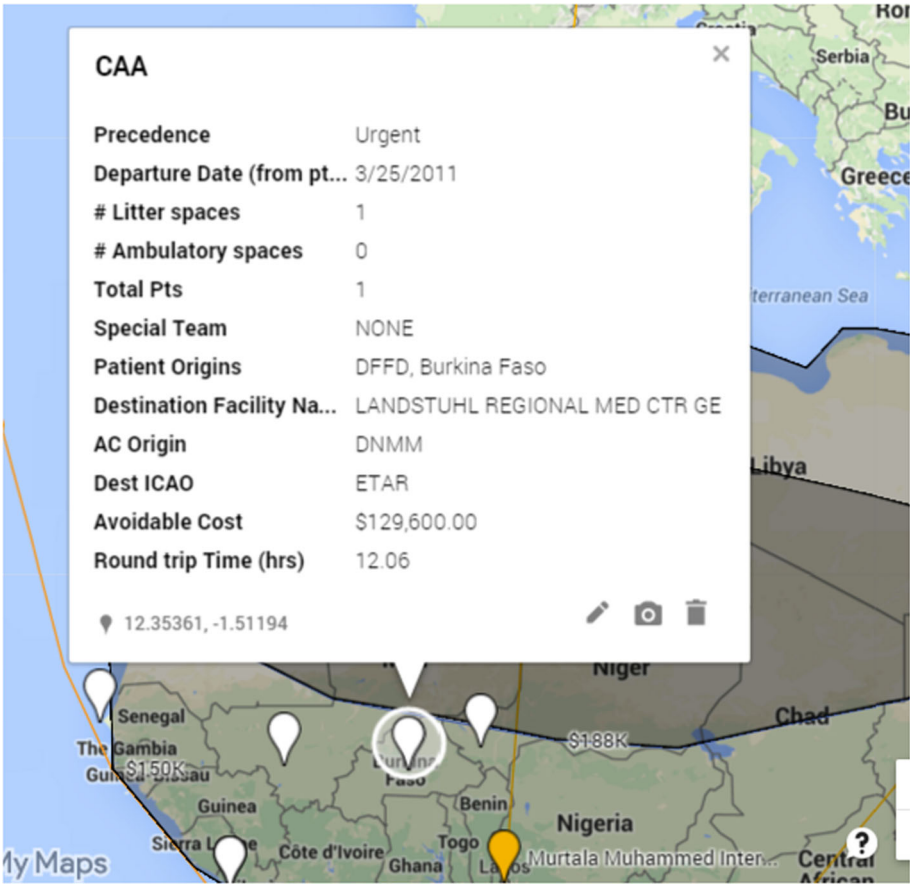

Fig. 3 Patient movement information example

best opportunity to select the most appropriate aircraft. Further analysis revealed that 23% of Routine missions used a C-130 military aircraft when a smaller C-21 was more appropriate. In addition, 35% of Routine missions used civilian air ambulances when either a C-21 or C-130 was more appropriate. Of the Urgent missions, 17% of them flew on a C-17 although a civilian air ambulance was more appropriate. In addition, we found cost savings on 40 of the 46 missions identified as a sub-optimal aircraft choice. For the other six missions, our decision tree recommended a more expensive aircraft as the best choice due to the reach and speed capabilities of the aircraft. The total savings were $2.6 million over 5 years. This is a savings of $520 thousand each year, which is approximately $15 thousand per mission.

4 Discussion and conclusions

4.1 Contributions to literature and practice

This research provides two important contributions that have not received necessary attention in the humanitarian logistics literature. First, we illustrate how a simple, yet effective, analytic method and open-source imagery tools can offer significant value to humanitarian

operations. With the majority of humanitarian operations modeling being narrow in scope and burdensome in computational feasibility (Özdamar and Ertem 2015), our approach adds breadth and computational simplicity to the analytic modeling approaches employed. Furthermore, with the intuitive nature of decision trees, the approach is interpretable by front-line humanitarian logisticians and planners and aligns to their particular decision making process. In addition, the deployment of this method in a self-service analytic mapping tool places the data discovery, visualization, and decision tree capability in the hands of practitioners. By enabling the front-line humanitarian planners and decision makers with a self-service analytic tool, the analytic approach becomes more reproducible, consistent, and available on-demand (Nijkamp et al. 2009). Furthermore, the analysis-to-decision cycle time is minimized. Combined, the problem addresses in this study helps to provide the necessary relevance of analytics and decision-making frameworks developed in academic research for practitioners (Hazen et al. 2016). Furthermore, with the increase in humanitarian disasters, this research addresses the growing need for the research community to provide such decision support tools (Coles et al. 2017).

Second, this research effort illustrates the challenges of drawing a clear picture from disparate data sources in the humanitarian operations domain. Organizations have been capturing data for years and conducting analyses to gain market intelligence. Historically, the orchestration of organizational data was relatively simple because data was well-structured and processed, and stored in traditional data management applications. Within the humanitarian domain, many large humanitarian organizations are just recently implementing traditional Enterprise Resource Planning (ERP) systems (Kovács 2014). However, in the age of "Big Data", a variety of external data sources are becoming available for humanitarian analytic efforts. Social media data are being collected to identify discussion topics, locations, and sentiments regarding emergencies. For example social media data has been collected and analyzed to assess refugees passing through Europe (OCHA, UN 2016). During the 2012 Hurricane Sandy disaster in New York, social media content was captured to assess both demand and supply of relief items (Imran et al. 2013). Leveraging these disparate data sources creates significant challenges.

This research illustrated its own challenges with collecting, storing, processing consolidating and operationalizeing data from disparate sources. All the data required to address our analytic problem was external to the main decision-making organization and stored in a variety of ways (i.e. restricted databases, online application program interfaces, individual Excel workbooks, regulation PDF reports). Collecting, storing and processing this data footprint creates significant challenges for organizations with traditional ERP systems since they were not developed to handle the scale, scope and disparity of today's data sources (Chen and Zhang 2014; Kambatla et al. 2014). Thus, significant time and effort was required to obtain access to, collect, and trust our data. Some data were encoded in non-intuitive ways and required us to consult data owners and stewards and reference master documents. This is not an entirely new challenge as it reported that approximately 80% of data analysis is spent on the process of cleaning and preparing data (Dasu and Johnson 2003; Wickham 2014). However, to date, little if any research in the humanitarian logistics domain has emphasized this challenge. Thus, the challenge humanitarian logisticians will continue to face is that they will become richer in data but poorer in information.

4.2 Limitations and future research

This research was developed to establish an analytic tool that provides humanitarian planners a self-service data discovery, visualization, and decision assessment capability for patient

evacuation in Africa. As with any research, limitations exist. We note that our framework's scope is quite specific to the decision context. Although our approach illustrates how future research can create more practitioner relevant capabilities and tools, the more specific these tools become the less external validity they tend to have. As such, future research could work to develop tools that generalize the same capabilities but are not specific to the military or medical domains. Such a tool could provide evacuation decision trade-offs for personnel regardless of the humanitarian disaster (e.g. medical, natural, man-made), incorporate more transportation options (e.g. military, civil, non-profit, for-profit), or cover more locations.

However, this introduces another limitation. As illustrated, the data collection and cleaning process was cumbersome. However, the current tool merely corralled the available data at the time of the research. To continue providing the most relevant information, future data collection will be required. Thus, future research could automate this process so that these data sources continually feed to a central database and provides the self-service tool with real-time data. This becomes more challenging as future research generalizes the tool to more disaster domains and incorporates more participating organizations and countries. Each of these advancements will incorporate new and different data sources requiring greater data collection, cleaning, and processing efforts. Thus, future reseach must address how humanitarian organizations can advance from their current ERP approach for data management to one that fosters the current big data environment with disparate and disjointed data.

Lastly, the long admired Occam's Razor is usually interpreted to mean that simpler is better. Unfortunately, in prediction, accuracy and simplicity (interpretability) are often in conflict. Decision trees rate high on interpretability. Furthermore, for our decision process decision trees improved the decision effectiveness and efficiency as compared to the status quo process and, potentially, could have saved the U.S. Air Force $2.6 million on past decisions. However, in the spirit of George Box, "all models are wrong, but some are useful" (Box 1976), the authors readily admit that more advanced analytic approaches may outperform the accuracy and effectiveness of our decision tree approach. Future humanitarian research should work to develop and compare multiple modeling approaches that span the spectrum of accuracy and simplicity as "assorted views of the same underlying data are often helpful. Multiple portrayals may reveal multiple stories, or demonstrate that inferences are coherent, or that findings survive various looks at the evidence in a kind of internal replication" (Tufte 2006). Thus, humanitarian logistics research would be well served to continue advancing the breadth of analytic tools available for practitioners.

4.3 Concluding remarks

Analytics need not be complex to offer significant value to humanitarian logistics operations. The decision-making context and feasibility of implementation should guide the solution. As humanitarian efforts and decisions grow, humanitarian logisticians require a larger set of tools that span the spectrum of accuracy, simplicity, computationally feasible, and practically relevant.

Our decision tree approach fills a current gap in humanitarian analytic approaches that are simplistic and computationally feasible. Furthermore, embedding this approach in a self-service analytic tool enabled us to place the data discovery, visualization, and decision tree capability in the hands of the practitioners. The tools offer planners an intuitive, quick method to improve the medical care delivered to military people deployed on the African continent. These tools can easily be adapted to other contexts. However, a growing challenge for humanitarian logistics analysts, and possibly the most significant obstacle, may lie in the ability to access reliable data. While possibly plentiful, transactional data on humanitarian

contingencies tends to arrive in disjointed chunks—some as Internet download files from widely known systems, but other datasets as spreadsheets passed from individuals or small organizations with limited resources or public awareness. This research helps to shine a light on the challenges of drawing a clear picture from disparate data sources in the humanitarian operations domain.

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